

Lessons Learned Building Low-Cost DIY Tactile Graphics and Conducting a Tactile Drawing Club in Colombia During COVID-19

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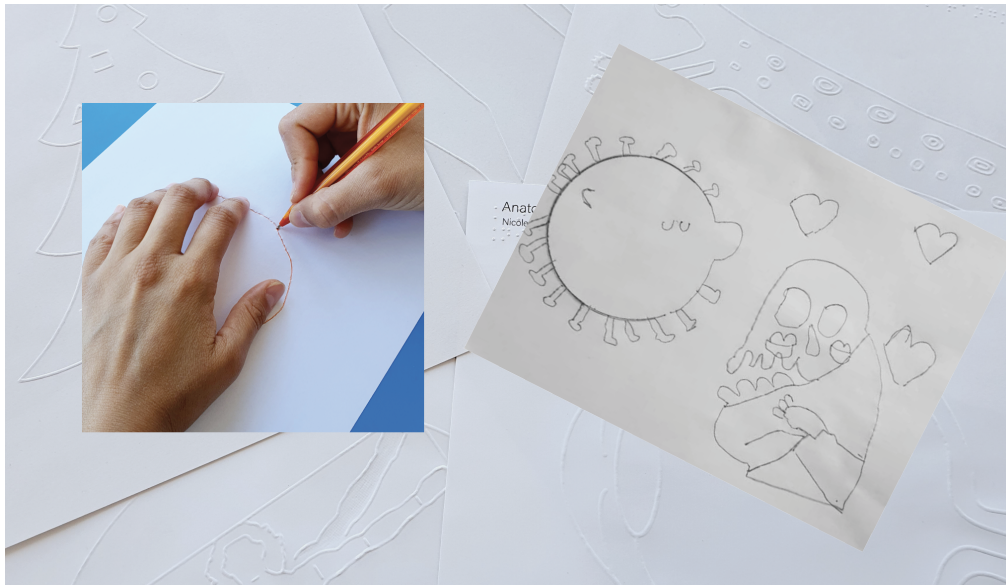


Figure 1: Materials used in the tactile drawing club. In the background are photographs of low-cost tactile graphics created using the stencil embossing contour line technique. The image on the left shows the low-cost tactile drawing technique, while the right one shows one of the drawings made by a blind participant of the tactile drawing club

ABSTRACT

Perceiving images and drawing are fundamental parts of human life, and thus access to them should be a universal right. However, there is a large breach for people with visual impairments to access diverse graphics, let alone drawing. There are several techniques of tactile graphics, such as swell paper, Braille embossing, and thermoform that help to alleviate this gap. However, in developing countries, the high cost and lack of access make them impractical. In this work, we describe our experience improving access to tactile graphics and drawing in Colombia. We created low-cost, effective and efficient, tactile graphics and drawing techniques that improve

on current solutions. These techniques were created from the best practices of two projects adapting pieces from the Colombian art heritage [52, 53] for blind and visually impaired people. They were then applied to a third project: running a virtual tactile drawing club with blind and visually impaired participants in the middle of the COVID-19 pandemic. The lessons learned from these experiences are presented in this paper with the hope they can help the community democratize access to tactile graphics.

CCS CONCEPTS

• **Human-centered computing** → Accessibility systems and tools; Empirical studies in accessibility; • **Social and professional topics** → People with disabilities.

KEYWORDS

Tactile Graphics, Tactile Drawing, DIY, Blind Embossing, Tactile, Home, Visual Impairments, Blind, Non-visual Interaction, Remote Participation

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1 INTRODUCTION

In developing countries as Colombia, access to tactile graphics for blind and visually impaired people is scarce [17]. To make the situation worse, some tactile graphics production methods [22] such as thermoforming and Braille embossing tend to be too expensive when considering the buying power of the citizens. According to the Colombian government, for 2020 about 42.5% of the population fell behind the line of monetary poverty [16], which measures the percentage of the population with income below the minimum monthly income defined as necessary to cover their basic needs.

Because of costs, technological solutions are seldom used, but parents and teachers of the visually impaired (TVIs) resort to hand-tooling and collage techniques to make do; On the other hand, official institutions, libraries, and schools tend to offer services to blind and visually impaired people using thermoforming and Braille embossing, but these tend to be mostly for text and not graphics. Many libraries in Colombia that have a Braille embosser, were not aware it was possible to use them for tactile graphics generation or how to do it.

The landscape for tactile drawing by blind and visually impaired people in Colombia is even more scarce. Most blind people we interacted with didn't believe they could draw. However, the Instituto Nacional para Ciegos INCI (National Institute for the Blind in Spanish) sells a positive-drawing board, which consists of a table and a plastic mesh surface on top of which people can draw using regular paper and crayons [12].

Given all these restrictions, the few tactile graphics and tactile drawing boards available for education and entertainment have to be commonly shared between individuals and returned to the institutions to be reused multiple times. This model tends to work better at larger metropolitan areas such as Bogotá, but is less effective in smaller cities and rural areas where transportation from the capital can take multiple days.

During the COVID-19 pandemic, even in developed countries such as the United States and Canada, education institutions had to shift to remote learning. This led to less availability of tactile graphic materials compared with what used to be available at the schools [44]. In developing countries, the situation was much worse. With the start of the pandemic, most Colombian cities entered a lockdown. Blind and visually impaired students who relied on sharing the same tactile images in a shared physical space such as a library, found themselves quarantined at their homes, with little to no access to tactile graphic materials.

On the importance of drawing. The earliest evidence for a drawing in the archaeological record dates to 73,000 years ago [1]. Nowadays, drawing can play a significant role in the growth and development of human thinking and education [9]. Although the sense of sight has been given a leading role in drawing [19], today we know that the part in the brain that deals with space, shape, and

form can be reached through vision as much as through touch [36, 38]. And despite this, many blind and visually impaired people are unaware that they can stimulate this part of the brain through tactile drawing. This paper presents a low-cost alternative for people who are blind and visually impaired to draw with the hopes that it can help alleviate this problem.

This work also presents the lessons learned with our efforts to improve the access to tactile graphics and drawing in Colombia. We first describe the experiences generated adapting 42 art pieces [52] from the Colombian National Library [2], using state-of-the-art techniques including Braille embossing (low and high resolution) and swell paper. The results of this project showed that despite the known benefits of these techniques, their costs made them unfeasible for Colombia. To address this, a second project was conducted using an improved, low-cost hand-tooling tactile graphic technique to adapt 12 pieces from the local artist Lucy Tejada [53]. This technique was improved even further by using a low-cost cutting machine for generating stencils that streamlined the production of tactile graphics. Using this as well as a novel tactile drawing technique, a remote tactile drawing club for the blind was created to address the lack of good graphical materials during the COVID-19 stay-at-home orders. The club has been a success over the last 15 months, since October 2021, meeting remotely once a month to appreciate relevant tactile graphics and practice tactile drawing from home.

The main contributions of this work are:

- An improved method for small production of tactile graphics for blind and visually impaired people from home, which uses a low-cost cutting machine for embossing contour lines. The advantages of this method include the low-cost DIY (do it yourself) nature, high tactile quality, and uniformity in the tactile graphics produced. One disadvantage of the method is that it can be more time-consuming than high-cost solutions such as swell paper.
- A low-cost tactile **drawing** technique for blind and visually impaired people, which can be used at home without many requirements.
- The lessons learned and best practices derived from two projects adapting art pieces from Colombian artists and running a remote tactile drawing club for blind and visually impaired people.

The rest of this paper starts by describing a sample of the relevant related work for tactile graphics and tactile drawing techniques from the state of the art. We then present two experiences using tactile graphics for adapting pieces of art from Colombian artists for blind and visually impaired people, one before the pandemic and the other at its beginnings. It follows with an introduction of the remote tactile drawing club and the descriptions of the proposed low-cost DIY tactile graphics production and drawing techniques. Some lessons learned are described from this experience, and the users' satisfaction with the tactile drawing club is reported, to finish with the future work and some conclusions of our work.

2 RELATED WORK

This section describes the state of the art of tactile graphics, tactile drawing, and remote event participation for blind and visually impaired people.

2.1 Tactile Graphics

A tactile graphic is a simplified, two-dimensional adaptation of the world, understandable with the sense of touch. Blind and visually impaired people should have a concept development about the three-dimensional nature of the object and their spatial relationship to have a better understanding of tactile graphics [28, 32].

There are standards and guidelines for tactile graphics design [8, 20, 29, 33, 45, 46] and techniques for its production according to the technology available in the market [18, 21, 22, 27, 50]: hand-tooling, collage, sewing, swell or capsule technology, blind embossing, Braille embossing (high and low resolution), silkscreen, thermoform, 3D printing, tactile displays, 2.5D printing, ultraviolet (or UV) printing, and raised-line drawing (tactile drawing) among other. However, the cost of these techniques create a lack of access in countries like Colombia were acquiring a basic Braille embosser costs more than 30 months of minimum wage (i.e. *embosser price: US\$7270* [49] * (*dollar price: COP\$3782.10*)/(*minimum wage: COP\$908526*) = 30.26).

There have been some less expensive mass-production techniques in developing countries as India using 3D printing and thermoform [26], but 3D printers are still out of reach for most Colombian households.

Blind embossing. This is an old technique to create images in relief with paper that has been used for blind and visually impaired people with stencil embossing of shapes [10, 11] and making art and the history of art accessible [13]. These DIY techniques can be significantly lower cost but are difficult to produce in mass and can lead to tactile quality issues.

Do-It-Yourself (DIY) movement. With the prevalence of normal difficulties involved in preparing tactile graphic [48] the Do-It-Yourself DIY movement has been an opportunity for parents and teachers of the visually impaired TVIs [23, 24, 31, 41] especially for small quantities production of tactile material.

2.2 Tactile Drawing

It is known that blind and visually impaired people can participate in tactile drawing [4–7, 35, 36, 39].

Tactile drawing differentiates from other types of drawing in that it allows for tactile feedback while strokes are made. There are two types of tactile drawing, negative-drawing, and positive-drawing, the latter allows one to feel the strokes immediately on the same surface where they are made. There are different tools on the market to generate them [30, 42]: homemade and commercial tactile drawing boards, rubber pads, quick-draw paper, drawing laces boards, velcro lace drawing boards, 3D pens, tactile displays, and tactile draw applications among others. At the time of this writing, the cheapest drawing board costs *US\$29.95* + taxes in the US [14], which makes it still inaccessible for Colombia.

2.3 Remote Participation

Blind and visually impaired people, who primarily use screen readers, shifted to remote experiences because COVID-19 pandemic [15, 43, 44]. They reported using tools such as Zoom or Google Meet and receiving materials at their home. For our remote tactile drawing club, we used a similar approach, shipping by regular mail the materials (i.e. the silicone placemat for drawing, and the tactile graphic used), and holding the meetings using Zoom.

3 BACKGROUND: BEFORE AND DURING COVID-19

This section describes the landscape of tactile graphics for Colombia, as well as two projects adapting pieces from the Colombian artistic heritage for blind and visually impaired people. These projects served as a base for the tactile graphics production and tactile drawing techniques, as well as for the remote tactile drawing club.

In Colombia before the pandemic, access to tactile graphics was mainly offered by TVIs (teachers of the visually impaired) in schools. The main techniques used are collage and hand-tooling [12]. There are many experimental projects targeting tactile graphics that come from universities, researchers [25, 47] or artists, but these tend to have a one-time appearance in newspapers and social media. Moreover, these experimental projects usually don't meet the real-world requirements of parents and TVIs or lack the production/distribution channels to be used in the wild.

3.1 Sintiendo el Patrimonio Colombiano

The National Institute for the Blind INCI offers some thermoformed tactile graphics, but to the best of our knowledge, other tactile graphics techniques such as swell paper and Braille embossing are largely unknown. To address this, and in a similar way to other projects adapting tactile graphics of art in the Philippines [40], we created the project *Sintiendo el Patrimonio Colombiano* (Feeling the Colombian Art Heritage) [52]. The objective of this project, which was executed between 2019 and 2020, was to make accessible for blind and visually impaired people 42 art pieces from the archives of the Colombian National Library [2] using swell paper and Braille embossing (low and high resolution) [37] and tactile drawing. Braille embossers were selected because the Colombian Government has endowed these machines to a wide network of local Libraries and other technological spaces called *Vive Digital*. Due to unnecessary bureaucracy, these Braille embossers are difficult to access for public use. Each Library tends to have only one of these machines, and they were commonly placed in the middle of a reading room without any sound protection. Given that these machines are particularly noisy, this draws them unusable without disturbing readers. People in charge of the machines didn't know how to use them for tactile graphics generation, and in many cases were reluctant to let us use them. In some cases, the printers were unused because of a lack of access to the proper paper.

In the case of swell paper, we wanted to introduce the technology to the country as it is mainly unknown¹. However, getting access to the paper is very difficult, as one sheet of swell paper costs about thirty times more in Colombia than in the US after considering

¹To the best of our knowledge, this project imported the first swell paper machine to Colombia, which is now available at the National Library

importing fees and adjusting for spending power², this is if you can import it yourself because there isn't a vendor that sells them in the country.

As part of this project, many training sessions were held teaching TVIs and Library personnel how to create tactile graphics using the different technologies. The sessions also included a section on tactile drawing, using a low-cost silicon placemat, a ballpoint pen, and regular paper, which proved to be a very cost-effective alternative at around US\$2 (when purchased in bulk) compared to a similar product as the Sensational BlackBoard Mini that costs US\$29.95 [14] in the US (importing and shipping fees would need to be added).

3.2 Lucy Tejada Táctil

Once the pandemic had started, the country went into a nationwide quarantine. Hence, we started to think of ways to create tactile graphics from home, with safe distribution and interaction for the blind and visually impaired people in their homes. To address this, we created the *Lucy Tejada Tactile* [53] project, whose objective was to adapt for the blind 12 art pieces from the Colombian female artist Lucy Tejada[51], as a commemoration of what would have been her 100th birthday. The tactile graphics were then safely distributed via regular mail to the homes of about 20 blind and visually impaired participants, for a later virtual meeting using Zoom, where the pieces were thoroughly verbally described so the participants could understand the adaptation.

For this project, we experimented using hand-tooling with regular paper over padded leather (i.e. such as the hardcovers of some books and agendas). We also experimented with different materials to produce textures, e.g. with plastic meshes. We concluded that the technique produced very high-quality tactile graphics that our participants really enjoyed. The mail distribution and remote meetings using Zoom also worked well, despite some common hiccups with delayed mail and technology issues. The downside of the production technique was that it was too time-consuming to prepare material for 20+ people. This situation led us to develop a new technique that we called the Stencil Embossing Contour Line Technique that we describe next.

4 STENCIL EMBOSSEING CONTOUR LINE TECHNIQUE

We present a new DIY technique for developing tactile graphics at home that we call **stencil embossing contour line** as seen in Figure 2. We first create an acetate stencil using a low-cost cutting machine outlining the selected tactile graphic. The stencil can then be used to streamline the production of copies of the chosen tactile graphic. We normally produce 20 copies of each piece, using relief paper (Braille paper - bond paper 115 g, letter size). This technique produces consistent copies, with high-quality raised lines, and it doesn't need any special wrapping when shipped by postal mail. This technique is an improvement of the blind embossing technique in special to the stencil embossing of shapes used for blind and visually impaired people [10, 11]. We focus on the contour line of shapes because those simulate the strokes in a draw. Blind and

visually impaired people can follow the contours on the tactile graphic and try to replicate them in their creations easily and with more clarity.

4.1 Instructions for using the stencil embossing contour line technique

This technique is composed of two parts, creating the stencil, and then using it for hand embossing.

4.1.1 How to create the stencil. **Materials:**

- (1) Acetate film (matte color), thicknesses .010" (e.g. Grafix Dura-Bright).
- (2) Clear self-adhesive film (e.g. ConTact).
- (3) A home cutting machine and their tools (e.g. Cricut).

Steps:

- (1) Create the design in software like Illustrator, only tracing the contour line, stroke: 4pt. Then select all the lines and click option Object / Path / Outline Stroke³, with "Shape Builder Tool" joining all segments, save it in a .svg file. This file is the stencil.
- (2) Upload the stencil .svg file in the software of the cutting machine.
- (3) Prepare an acetate film to be cut in the Cricut or cutting machine.
- (4) Once cut, put on a clear self-adhesive film piece, and then peel off the adhesive film and acetate together from the surface of the cutting machine.
- (5) Using a sharp tool, carefully peel off the contour line from the acetate and adhesive film.
- (6) Add baby powder in the ditch of the stencil to eliminate any remaining adhesiveness.

4.1.2 How to Use the Stencil for Hand Embossing. **Materials:**

- (1) Stencil.
- (2) Braille paper - bond paper 115 g.
- (3) Embossing stylus.
- (4) Braille slate and stylus.
- (5) Tracing wheel.
- (6) Masking tape and scissors.

Steps:

- (1) Attach with tape a Braille paper to the stencil on the side of the ditch. Try using just enough tape to temporarily keep the two pieces together.
- (2) Place the stencil and Braille paper on a rigid surface, which generates enough contrasts with the stencil (a lightbox would be ideal, but not required). Using an embossing stylus trace through the ditch.
- (3) Carefully separate the Braille paper from the stencil.
- (4) On the Braille paper, write the title with the Braille slate and stylus and make the orientation mark with the tracing wheel (two diagonal lines in the top right corner, remember to flip the sheet to do it).

²29.98 = $\frac{1 \text{ sheet of swell paper in Col.} \approx \text{US\$3} \cdot \text{dollar} : \text{COP\$3782.10}}{\text{a day of min. wage in Col.} : \text{COP\$30284.2}} / \frac{1 \text{ sheet of swell paper in Cal.} \approx \text{US\$1.5}}{\text{a day of min. wage in Cal.} : \text{US\$15} \cdot 8 \text{ hours}}$

³Instructions for Illustrator 2021



Figure 2: Stencil embossing contour line technique

5 THE REMOTE TACTILE DRAWING CLUB

Using all the knowledge learned from the previous projects we started a Remote Tactile Drawing Club. This monthly activity was a way to teach and incentivize tactile drawing in Colombia. It also offered a social activity for blind and visually impaired people to gather safely amidst the pandemic. Figure 1 shows in the background examples of the tactile graphics generated for the club, as well as one of the drawings made by one participant. The call for participation in the club is sent via WhatsApp messages, because of its wide use in Colombia. The recipients of these invitations started from the same participants of the two previous projects [52, 53], and it has been expanding organically through word of mouth. The participants of the tactile drawing club are blind and visually impaired people (blind and low vision), with an age range of 17 to 54 years old. In the 15 months of the club, 27 blind and visually impaired people out of 9 cities in Colombia have participated. The club is not for profit and is completely free for all the participants.

The club has been meeting virtually using Zoom calls since October 2020, every second Saturday of the month for about one and a half hours. Before the club, they receive in their house via postal mail a copy of the tactile graphic of the masterpiece, which

will be replicated by tactile drawing in the club as can be found in Figure 3.

If there are new participants in the club, we start by explaining how to trace lines to generate a tactile response using a silicone placemat, regular paper, and a ballpoint pen. The participants talk about their experiences and best practices. Then we do an introduction of the theme of the masterpiece of the club and we do warming-up exercises oriented in the same theme. Finally, we replicate the masterpiece selected for that session of the club. During the club, participants describe their drawings and talk about their difficulties, achievements, and their points of view.

Images adapted. In Colombia, the Law 1680 of 2013 in its article 12, is an exception to copyright that allows the adaptation of material for blind and visually impaired people without the authorization of their authors and only the obligation to mention their names and the title of the work. This has allowed us to adapt the different masterpieces for the club as can be found in Table 1 using the stencil embossing contour line technique.

low-cost Tactile Drawing. Along with the tactile graphic copy, we also ship extra pieces of regular paper (bond paper 75 g, letter size) to be used in the tactile drawing activity. To enable this, we



Figure 3: Tactile drawings of the masterpieces by one of the visually impaired participants of the tactile drawing club

found that the cheapest (yet effective) material for drawing is a silicone placemat. Tactile drawing on these mats is possible with regular paper and any type of ballpoint pen. Its cost is around US\$2 per unit, and it allows for positive drawing. These placemats were given free to the participants in their first participation in the club.

6 LESSONS LEARNED AND CHALLENGES

Here are some of the lessons from our experience with the three projects:

- The cheapest tool for positive tactile drawing in the market is a silicone placemat, regular paper, and a ballpoint pen.
- The stencil embossing contour line technique described in Section 4 is one of the cheapest methods to produce a small set of tactile graphics copies while maintaining the quality.
- The method to learn by doing combined with reproducing a tactile masterpiece is very effective in the tactile drawing learning process.

Table 1: Tactile Drawing Club Masterpieces Adapted by Month

Month Year	Masterpiece Title	Masterpiece Author	Masterpiece Type	Pieces Shipped	Cities
Oct 2020	Monument to the student	Lucy Tejada	Monument	10	2
Nov 2020	Jaguar - Chiribiquete rock art	Jorge Mario Álvarez Arango	Photography	20	7
Dec 2020	Christmas tree	APH	Contour line drawing	18	5
Jan 2021	A Handmade Labyrinth	Maia Scott	Drawing	16	5
Feb 2021	Three Circuit Labyrinth	-	Drawing	16	5
	Sunflower	APH	Contour line drawing		
Mar 2021	Perseverance Rover	NASA	Illustration	17	6
Apr 2021	Is the third "peak" coming?	Julio César González, Matador	Editorial cartoon	18	7
May 2021	The Vitruvian Man	Leonardo da Vinci	Artwork	16	8
Jun 2021	Eye front view	Selman Design	Drawing	16	7
	Eye anatomy	Nicole Johnson	Stencil		
Jul 2021	Bull side and front view	Nicole Johnson	Drawings	12	6
	The bull - lithograph 11	Pablo Picasso	Artwork		
Aug 2021	Tokyo 2020 Paralympic Medal	Sakiko Matsumoto	Medal	14	6
	Pictograms Paralympic Games	Masaaki Hiromura	Drawings		
Oct 2021	Life	Philipp Carlucci	Comic	12	7
Dec 2021	Fireworks shapes	-	Drawings	12	7

- One of the keys for successfully running the tactile drawing club remotely was to be very descriptive in the whole process, both in the use of tools, as of the masterpieces to replicate.
- Doing warm-up tactile drawing exercises before perceiving the masterpiece seems to have helped participants better understand the theme of the piece.
- At different points of the exercise, participants were asked to share their drawings, commonly by enabling their camera and pointing the drawing to the camera. Holding a conversation about the participant's intentions helped to improve the drawings and consider different ways of representing the same concept.
- Selecting topics and masterpieces that were of interest to the participants was fundamental. This was achieved by using common events happening around the time of the club (e.g. the Paralympic games) or directly asking participants for theme ideas for the next club.
- Every club session should have a script with the background story of the tactile masterpiece to help to provide context, increase the meaning, and to make it more memorable for participants.
- The participants need to have experience in the use of remote technologies. It is fundamental that they know how to control the microphone, camera, etc.
- It is important to figure out the average shipping times of the postal service used, to guarantee that the materials arrive on time for the club session.
- Identify a reliable communication channel to contact the participants, e.g. WhatsApp in our case. When choosing this channel, take into account the participant's familiarity with it.

- Create a secure environment where mistakes are necessary and desired for learning. Aim to make practicing the best incentive to acquire better drawing skills.

6.1 Remote Experience with Blind and Visually impaired People

Due to the COVID-19 pandemic, blind and visually impaired people shifted to remote experiences. In the tactile drawing club the challenges were:

- To have uniform tactile graphics for all the participants with Braille labels and orientation guides.
- To have a distribution system for all the country on time for the club.
- To properly use microphones and cameras in Zoom with all participants during the club.
- To be extremely clear in the description of the tactile tour of the piece of the club.
- To create a deep and meaningful context with the description of the piece of the club.
- To create a participatory environment to try, fail, achieve, learn, and keep going.

7 DRAWING CLUB PARTICIPANTS' SATISFACTION EVALUATION

The blind and visually impaired users of the club shared their experiences in a virtual public event [3] as a celebration of one year of the Tactile Drawing Club on November 2021.

In addition, we collect feedback with an anonymous satisfaction survey conducted at the end of the year 2021. The survey was sent to every person who participated at least twice in the club, 23 in

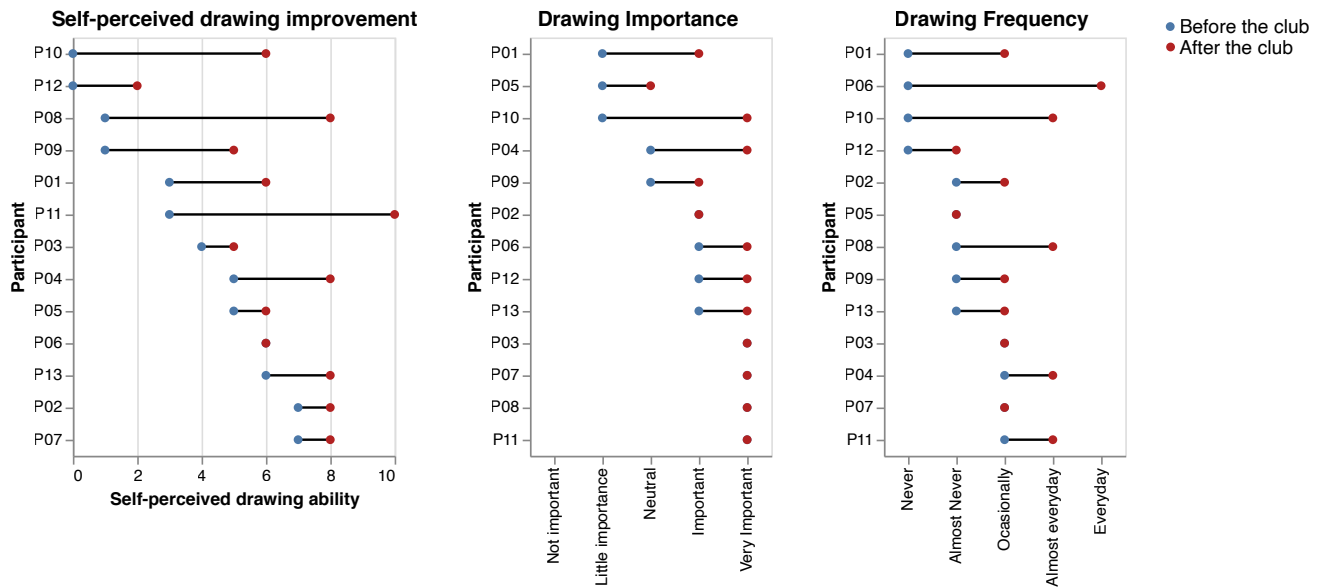


Figure 4: Participant's survey responses. All participants reported that they had increased or equaled their perceived drawing ability, the importance of drawing, and the frequency they drew

total. Out of these 13 people completed the survey. The survey included 21 questions including demographics, the participant's perception of the importance of drawing before and after the club, their perceived drawing skills before and after, among others⁴.

Out of the 13 participants who responded 12 identified as blind (2 born blind), and 1 as acquired low vision. The responses came mainly from the recurrent participants: 6 participants reported having attended 10-13 sessions, 2 6-9 sessions, 3 3-5 sessions, and 2 people participated in 2 sessions.

Participants were asked to rate their self-perceived drawing ability on a scale of 0 (no ability at all) to 10 (expert ability) before and after the club. All responders reported increases in their self-perceived drawing ability, as shown in Figure 4 on the left. The average increase was 2.92 (std. dev. 2.39). In a similar way, participants reported an increase in the perception of the importance of drawing and the frequency they drew. Everyone but one reported drawing to be important or very important in their lives (Figure 4 middle). In terms of the frequency of drawing, 10 participants reported increasing the frequency of their drawing, 3 kept it the same amount. 4 participants even reported improving from never drawing, two of them to drawing almost everyday or everyday.

These results confirmed the relevance of these kinds of activities and showed that when given the opportunity, blind and visually impaired people are interested and value drawing. As an anecdote, one participant shared freely at the end of one of the clubs, that the drawing technique had allowed her to get closer to her daughter by enabling her to discuss visual concepts of her homework using drawings. Another participant mentioned that he had used it to interact with his students using graphical concepts, both with him

and the students using the placemat to draw. These stories reflect on the impact the tactile drawing club has on its participants and motivate us to continue our work.

8 CONCLUSIONS AND FUTURE WORK

In order to increase the adoption of tactile drawing and tactile graphics, a low-cost, yet effective technique to produce them is required. This is particularly true for developing countries, where blind and visually impaired people have very low spending power. With that goal in mind, we developed three projects on tactile graphics and tactile drawing in Colombia. This paper presents the lessons learned from those projects, as well as low cost, DIY, tactile graphics generation, and tactile drawing techniques. We present these results with the hope they can be useful for the community.

The following are some further considerations of our work:

How to scale up the tactile drawing club. As it is, the tactile drawing club depends on us producing the materials, distributing them, and organizing all the logistics. However, just as special interests "meetups" are held around the world, tactile drawing clubs could be organized in a similar, and decentralized way in other countries. Right now, many organizations that serve blind and visually impaired people could amplify their services through a tactile drawing club.

Accessibility of the techniques. The tactile drawing technique is accessible. However, the hand embossing step of the Stencil Embossing Contour Line Technique (tactile graphics generation), isn't in the way presented in the paper. It requires a sighted person manually tracing the stencil to generate the tactile graphic, using a lightbox or something else that generates enough contrast. In order to test the feasibility of making this step more accessible, in the tactile drawing club of December 2021 we sent a Christmas tree

⁴The complete survey can be seen in <https://forms.gle/bL9daWLmwW6FcDvt8> in Spanish

stencil similar to the one used in the Stencil Embossing Contour Line Technique. Blind and visually impaired participants were able to use it with a simple modification in the order of the elements: they used it with regular paper and Foamy (stencil, paper, and foamy in that order top-down). Where in the production technique with sighted people, we use Braille paper on top of the stencil over a rigid surface. This result is encouraging and suggests that with more work the Stencil Embossing Contour Line Technique can be made more accessible.

Reproducibility of the Stencil Embossing Contour Line Technique. Thanks to a collaboration with Nicole Johnson, she tested the technique above described with her own designs. She also created and adapted some of the digital pieces for the tactile drawing club of June, July, and October. Moreover, she adapted 70 of her designs with this technique. The designs are available on the BTactile web page [34].

8.1 Future work.

The process of embossing by hand when producing the masterpieces to be sent to the club participants is a laborious one. Home press machines will be tried to alleviate this burden. High contrast color printing could be tried to improve tactile pieces for people with low vision. Finally, the stencil embossing contour line technique should be compared with other low-cost and DIY techniques to establish the advantages and disadvantages of each one.

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REFERENCES

- [1] 2018. The earliest known drawing in history sends a message through 73,000 years. *Nature* 561, 7722 (sep 2018), 149–149. <https://doi.org/10.1038/d41586-018-06657-x>
- [2] 2021. Biblioteca Nacional de Colombia [Colombian National Library]. <https://bibliotecanacional.gov.co>
- [3] 2021. Conversatorio Club de Dibujo Táctil [Tactile Drawing Club Discussion] - YouTube. <https://www.youtube.com/watch?v=k5W0lgrRVDs>
- [4] Elisabeth Salzhauser Axel, Nina Sobol Levent, and Editors. 2003. *Art beyond sight: A resource guide to art, creativity, and visual impairment*. American Foundation for the Blind. <https://books.google.com/books?hl=en&lr=&id=B4ioCFic7m0C&oi=fnd&pg=PA11&dq=art+beyond+light&ots=YngA1dGyFe&sig=U4LjUcnjabUD9v5SroqKncdV5hY>
- [5] Lola Bardisa. 1992. *Cómo enseñar a niños ciegos a dibujar [How to Teach Blind Children to Draw]*. ONCE. <https://www.foal.es/es/biblioteca/como-ensenar-ninos-ciegos-dibujar>
- [6] Jens Bornschein, Denise Bornschein, and Gerhard Weber. 2018. Blind Pictionary. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems*, Vol. 2018-April. ACM, New York, NY, USA, 1–4. <https://doi.org/10.1145/3170427.3186487>
- [7] Jens Bornschein, Denise Prescher, and Gerhard Weber. 2015. Collaborative creation of digital tactile graphics. In *ASSETS 2015 - Proceedings of the 17th International ACM SIGACCESS Conference on Computers and Accessibility*. Association for Computing Machinery, Inc, 117–126. <https://doi.org/10.1145/2700648.2809869>
- [8] Braille Authority of North America. 2010. Guidelines and Standards for Tactile Graphics. <http://www.brailleauthority.org/tg/>
- [9] Margaret Brooks. 2009. Drawing to Learn. *Making Meaning* (2009), 9–29. https://doi.org/10.1007/978-0-387-87539-2_2
- [10] Jan Carrol. 1998. Stencil-Embossed Tactile Images. *RE: view*, 30(3) (1998), 127–undefined. <https://www.proquest.com/openview/5561117ad635d23c1e0ff998ed300f95/1?pq-origsite=gscholar&cbl=48930>
- [11] Jan Carrol. 2001. Crafty Graphics Stencil Embossing Kit Guidebook. <https://sites.aph.org/files/manuals/7-08844-00.pdf>
- [12] Marta Emilia Castro Nungo and Elsa Marcela Del Castillo Sabogal. 2020. Material didáctico para estudiantes con discapacidad visual [Didactic Material for Visually Impaired Students]. <https://www.inci.gov.co/sites/default/files/cartillamaterialdidacticoparaestudiantescondv.pdf>
- [13] Hoëlle Corvest-Morel and Dorine in 't Veld. 2021. Library of tactile and high contrast images of art historical works. In *Tactile Reading*. Statped. <http://www.statped.no/tactile-reading-2021/recordings-29th-and-30th-april-2021/#universal-design>
- [14] Ann Cunningham. 2021. Sensational BlackBoard. <http://www.sensationalbooks.com/products.html#blackboard>
- [15] Ann Cunningham, Jenny Callahan, Abigale Jane Stangl, and Rishika Kartik. 2017. Tactile Art Club. <https://www.rishikastudio.com/tac>
- [16] Departamento Administrativo Nacional de Estadística – DANE [National Administrative Department of Statistics]. 2021. Pobreza monetaria [Monetary Poverty]. <https://www.dane.gov.co/index.php/estadisticas-por-tema/pobreza-y-condiciones-de-vida/pobreza-monetaria>
- [17] M. Bernardine Dias, Mohammed Kaleemur Rahman, Saurabh Sanghvi, and Kentaro Toyama. 2010. Experiences with lower-cost access to tactile graphics in India. In *Proceedings of the First ACM Symposium on Computing for Development - ACM DEV '10*. ACM Press, New York, New York, USA, 1. <https://doi.org/10.1145/1926180.1926193>
- [18] Polly K. Edman. 1992. *Tactile graphics*. American Foundation for the Blind. <https://books.google.com/books?hl=en&lr=&id=C7vq4-Th71AC&oi=fnd&pg=PR11&dq=tactile+graphics&ots=psecr1oFzS&sig=b1sSld9hWSZpNT4ajD9CXPNTcY8>
- [19] Betty Edwards. 2012. *Drawing on the Right Side of the Brain*. 1–284 pages. https://www.google.com/books/edition/Drawing_on_the_Right_Side_of_the_Brain/simRZwEACAAJ?hl=en
- [20] Yvonne Eriksson. 1999. How to make tactile pictures understandable to the blind reader - 65th IFLA Council and General Conference. In *65th IFLA Council and General Conference*. <https://origin-archive.ifla.org/IV/ifla65/65ye-e.htm>
- [21] Cheryl Fogle-Hatch, Ann Cunningham, and Matt Gesualdi. 2020. Staying in Touch: Addressing Concerns to Allow Tactile Exploration at Museums – American Alliance of Museums. *The American Alliance of Museums* (jul 2020). <https://www.aam-us.org/2020/07/01/staying-in-touch-addressing-concerns-to-allow-tactile-exploration-at-museums/>
- [22] John A. Gardner. 1996. Tactile graphics: an overview and resource guide. *Gale Academic OneFile* 3, No. 4 (1996). <https://go.gale.com/ps/i.do?id=GALE%7CA205493989&sid=googleScholar&v=2.1&it=r&linkaccess=abs&issn=10735127&p=AONE&sw=w>
- [23] Stéphanie Giraud and Christophe Jouffrais. 2016. Empowering Low-Vision Rehabilitation Professionals with “Do-It-Yourself” Methods. In *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. Vol. 9759. Springer Verlag, 61–68. https://doi.org/10.1007/978-3-319-41267-2_9
- [24] Stéphanie Giraud, Philippe Truillet, Véronique Gaildrat, and Christophe Jouffrais. 2017. “DIY” Prototyping of Teaching Materials for Visually Impaired Children: Usage and Satisfaction of Professionals. In *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. Vol. 10277 LNCS. Springer Verlag, 515–524. https://doi.org/10.1007/978-3-319-58706-6_42
- [25] Ricardo Gonzalez, Carlos Gonzalez, and John A. Guerra-Gomez. 2019. Tactiled. In *The 21st International ACM SIGACCESS Conference on Computers and Accessibility*. ACM, New York, NY, USA, 530–532. <https://doi.org/10.1145/3308561.3354613>
- [26] Richa Gupta, M. Balakrishnan, and P. V.M. Rao. 2017. Tactile Diagrams for the Visually Impaired. *IEEE Potentials* 36, 1 (jan 2017), 14–18. <https://doi.org/10.1109/MPOT.2016.2614754>
- [27] B. Hampshire. 1979. The design and production of tactile graphic material for the visually impaired. *Applied Ergonomics* 10, 2 (jun 1979), 87–97. [https://doi.org/10.1016/0003-6870\(79\)90059-0](https://doi.org/10.1016/0003-6870(79)90059-0)
- [28] Morton A. Heller, Melissa McCarthy, and Ashley Clark. 2005. Pattern Perception and Pictures for the Blind. *Psicología Universitat de València* 26, 1 (2005), 161–171. <http://www.redalyc.org/articulo.oa?id=16926112>
- [29] Ron Hinton. 1996. Tactile graphics in education. <http://www.ssc.education.ed.ac.uk/resources/vi&multi/hinton/index.html>
- [30] Laura Hospital. 2016. Art in Science: Tools for drawing for students with visual impairment. <https://www.perkinslearning.org/accessible-science/blog/art-science-tools-drawing-students-visual-impairment>
- [31] Amy Hurst and Jasmine Tobias. 2011. Empowering individuals with do-it-yourself assistive technology. In *The proceedings of the 13th international ACM SIGACCESS conference on Computers and accessibility - ASSETS '11*. ACM Press, New York,

- New York, USA, 11. <https://doi.org/10.1145/2049536.2049541>
- [32] Dorine in 't Veld. 2018. Concept-Building in Blind Readers with Thematic Tactile Volumes. In *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. Vol. 10897 LNCS. Springer Verlag, 185–192. https://doi.org/10.1007/978-3-319-94274-2_25
- [33] Dorine in 't Veld and Esther Rieken. 2015. Richtlijnen voor het ontwerpen van tactiele tekeningen op zwelpapier [Guidelines for designing tactile drawings on swell paper]. <http://www.tactieletekeningen.nl/wp-content/uploads/2016/01/Ontwerprichtlijnen-voor-zweltekeningen.pdf>
- [34] Nicole Johnson. 2021. BTactile.com: Stencil Designs. <https://btactile.com/?q=stencil>
- [35] John M. Kennedy. 1993. *Drawing & the blind: Pictures to touch*. Yale University Press. <https://tspace.library.utoronto.ca/handle/1807/1021>
- [36] John M. Kennedy. 1997. How the blind draw. *Scientific American* (jan 1997), 76–81. <https://www.jstor.org/stable/24993567>
- [37] Library of Congress - National Library Service for the Blind and Print Disabled (NLS). 2020. Braille Embossers. <https://www.loc.gov/nls/resources/blindness-and-vision-impairment/devices-aids/braille-embossers/>
- [38] Lora T Likova. 2012. Drawing enhances cross-modal memory plasticity in the human brain: a case study in a totally blind adult. *Frontiers in Human Neuroscience* 0, MAY 2012 (may 2012), 44. <https://doi.org/10.3389/FNHUM.2012.00044>
- [39] Al Maneki. 2013. The Dawn of the Age of Tactile Fluency: Let the Revolution Begin! *Future Reflections. Volume 32 Number 4* (2013). <https://nfb.org/sites/default/files/images/nfb/publications/fr/fr32/4/fr320406.htm>
- [40] Dennis A. Martillano. 2019. The Perceived Aesthetic Experience in the context of Tactile Art among Visually Impaired Individuals in The Philippines. In *Proceedings of the RSU International Research Conference*. 543–553. <https://rsucon.rsu.ac.th/proceeding/article/1571>
- [41] Samantha McDonald, Joshua Dutterer, Ali Abdolrahmani, Shaun K. Kane, and Amy Hurst. 2014. Tactile aids for visually impaired graphical design education. In *Proceedings of the 16th international ACM SIGACCESS conference on Computers & accessibility - ASSETS '14*. ACM Press, New York, New York, USA, 275–276. <https://doi.org/10.1145/2661334.2661392>
- [42] Annelise Qvale and Lisa Yayla. 2009. Husebys tegnepakke Veiledningshefte [Huseby's Drawing Kit Guidebook]. https://www.statped.no/globalassets/fagomrader/syn/dokumenter_syn/huseby_tegnepakke_veil_2009.pdf
- [43] Cecilia Rafie R. 2021. COVID-19 Pandemic: Threat or Opportunity for Blind and Partially Sighted Museum Visitors? *Journal of Conservation and Museum Studies* 19, 1 (mar 2021). <https://doi.org/10.5334/jcms.200>
- [44] L. Penny Rosenblum, Tina S. Herzberg, Tiffany Wild, Kathryn D. Botsford, Danene Fast, Justin T. Kaiser, Leanne K. Cook, Michelle A. C. Hicks, Jasamyn N. DeGrant, and C. Rett McBride. 2020. *Access and Engagement: Examining the Impact of COVID-19 on Students Birth–21 with Visual Impairments, Their Families, and Professionals in the United States and Canada*. Technical Report. American Foundation for the Blind. <https://www.afb.org/research-and-initiatives/education/covid19-education-research/access-engagement-study>
- [45] Round Table on Information Access for People with Print Disabilities Inc. 2005. Guidelines on Conveying Visual Information. <http://printdisability.org/guidelines/guidelines-on-conveying-visual-information-2005/>
- [46] Marco Schuffelen. 2002. On Editing Graphics for The Blind. <https://www.heardutchhere.net/grbl/grbl0.html>
- [47] Laura Lucia Serrano. 2018. INVISIBLES - Inclusive project of visual content creation related with endangered animals from Colombia. In *17 Festival Internacional de la Imagen*. Universidad de Caldas, Manizales, 494–497. https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q=+INVISIBLES-Inclusive+Project+of+Visual+Content+Creation+Related+with+Endangered+Animals+from+Colombia&btnG=
- [48] Linda Sheppard and Frances K. Aldrich. 2001. Tactile graphics in school education: perspectives from teachers. *British Journal of Visual Impairment* 19, 3 (sep 2001), 93–97. <https://doi.org/10.1177/026461960101900303>
- [49] Soluciones Integrales Ver [Integral Solutions See]. 2021. Impresora Braille - Index Everest-D v5 con Software de Transcripción a Braille Duxbury [Braille Embosser - Everest-D v5 Index with Duxbury Braille Transcription Software]. <https://ver.com.co/producto/impresora-braille-index-everest-d-v5-con-software-de-transcripcion-a-braille-duxbury/>
- [50] Abigale Jane Stangl. 2019. *Tactile Media Consumption and Production for and By People who are Blind and Visually Impaired: A Design Research Investigation*. Ph.D. Dissertation. University of Colorado Boulder. <https://scholar.colorado.edu/downloads/6w924c15s>
- [51] Wikipedia. 2021. Lucy Tejada Biography. https://en.wikipedia.org/wiki/Lucy_Tejada
- [52] Maria Fernanda Zuniga-Zabala and John A. Guerra-Gomez. 2019. Sintiendo el Patrimonio Colombiano [Feeling the Colombian Art Heritage]. <http://www.sintiendopatrimonio.co/>
- [53] Maria Fernanda Zuniga-Zabala, John A. Zuniga-Zabala, Angela A. Zabala, and John A. Guerra-Gomez. 2020. Lucy Tejada Táctil [Lucy Tejada Tactile]. <http://lucytejadatactil.co/>